

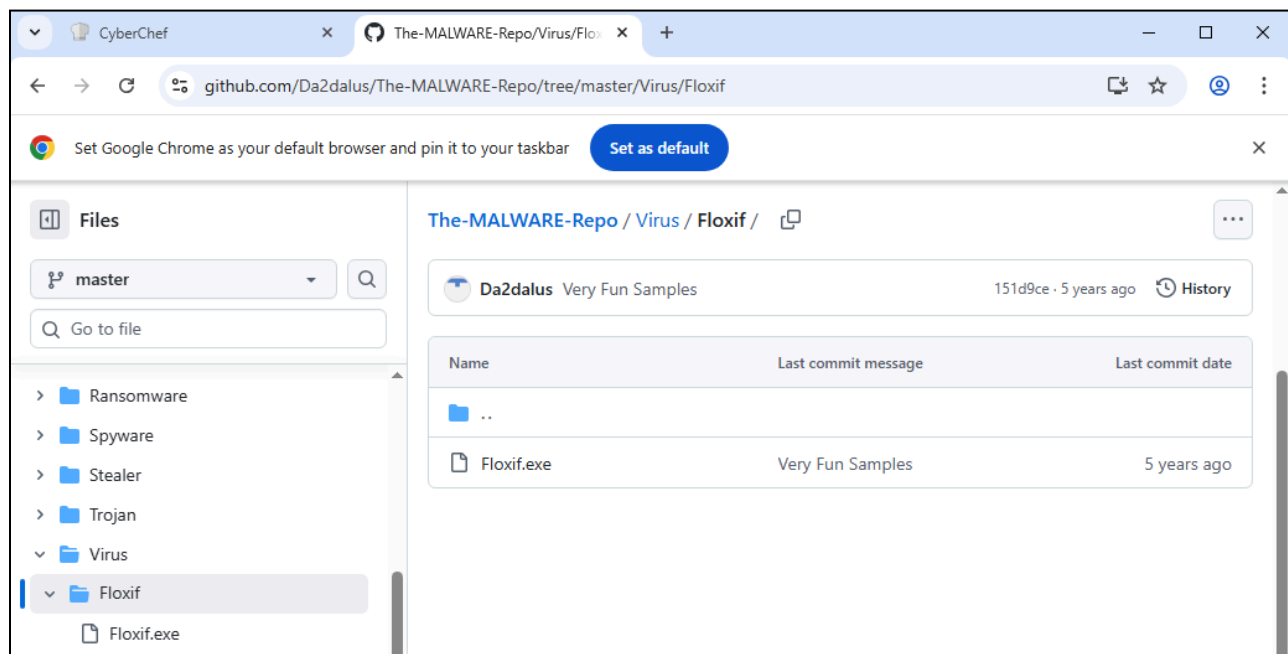
Malware Analysis A4

Objective:

A suspicious Portable Executable (PE) file is provided for static analysis. Using PEiD and PESTudio, analyze the file and answer the following:

1. Use PEiD to determine whether the executable is packed and explain why packing information is critical for interpreting imports and exports.
2. With PESTudio, extract the flagged APIs from the binary. For each key imported API, explain its functional significance (e.g., CreateRemoteProcess → process injection, WinExec → execution of external programs, SAsocketA → network communication).
3. Examine the export table (if present) and discuss how exported functions might indicate the malware's intended use (e.g., DLL-based persistence, function reusability by other malware).
4. Based on the observed imports and exports, infer the likely behavior profile of the malware (such as process injection, persistence, network communication, or file system manipulation).

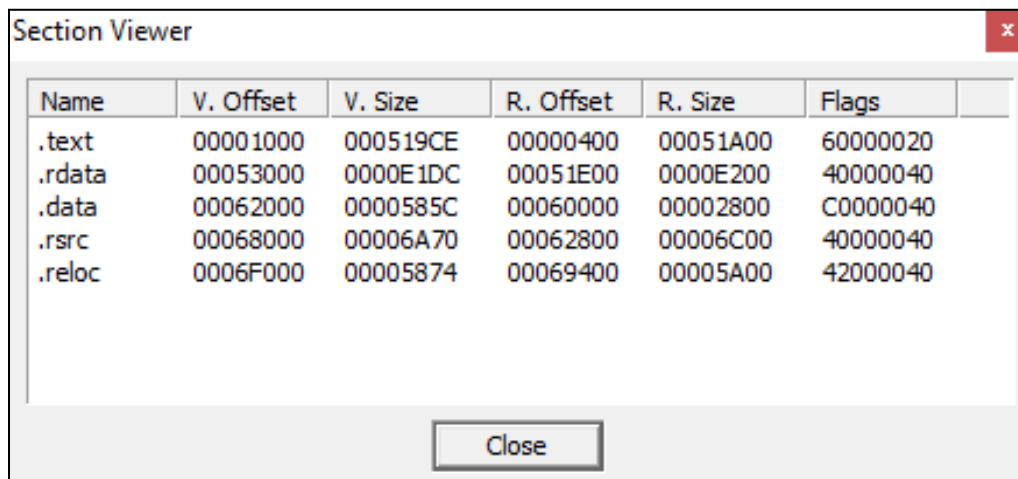
Target Malware : [Floxif.exe](#) from the Da2Dalus / Github



Tool Used: PEid

Steps to verify if the file is packed or unpacked:

- Identify the sections of the file - if it is a malicious file or packed file the section headings will be similar to UPX0,UPX1 etc..
- If the section headers are like .rdata, .data, .text etc., they are most likely to be unpacked.



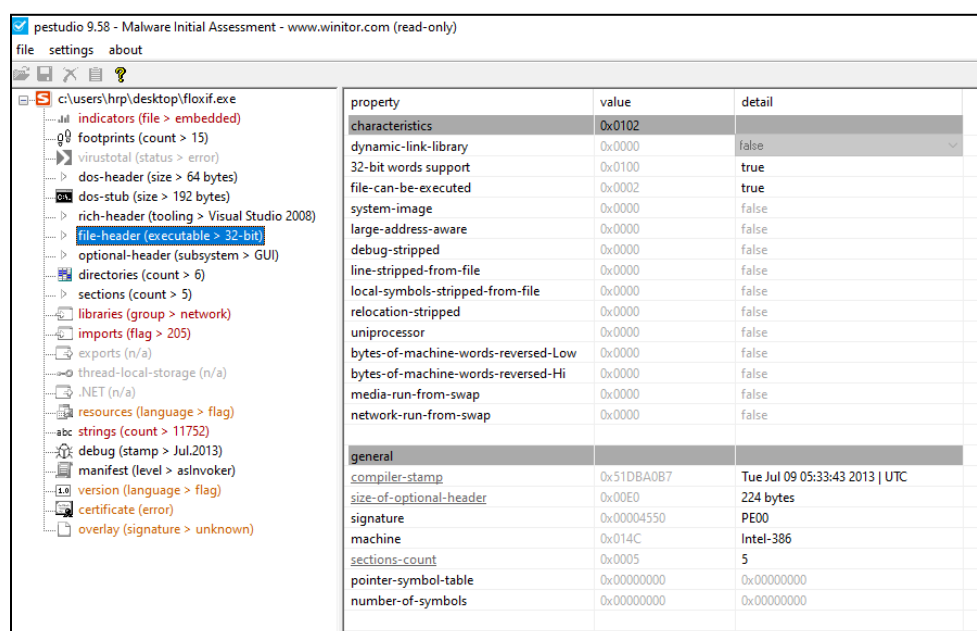
The screenshot shows the 'Section Viewer' window from the PEid tool. It contains a table with the following data:

Name	V. Offset	V. Size	R. Offset	R. Size	Flags
.text	00001000	000519CE	00000400	00051A00	60000020
.rdata	00053000	0000E1DC	00051E00	0000E200	40000040
.data	00062000	0000585C	00060000	00002800	C0000040
.rsrc	00068000	00006A70	00062800	00006C00	40000040
.reloc	0006F000	00005874	00069400	00005A00	42000040

At the bottom of the window is a 'Close' button.

- The following file is unpacked as we can see the .rdata, .data, .rsrc files

By checking the entropy we can check if it contains so many unwanted files or not, if the entropy is high - it's likely to have added more unwanted files to manipulate the OS.



The screenshot shows the PEStudio 9.58 interface. The left pane shows the file structure with 'file-header (executable > 32-bit)' selected. The right pane shows the 'property' table with the following data:

property	value	detail
characteristics	0x0102	
dynamic-link-library	0x0000	false
32-bit words support	0x0100	true
file-can-be-executed	0x0002	true
system-image	0x0000	false
large-address-aware	0x0000	false
debug-stripped	0x0000	false
line-stripped-from-file	0x0000	false
local-symbols-stripped-from-file	0x0000	false
relocation-stripped	0x0000	false
uniprocessor	0x0000	false
bytes-of-machine-words-reversed-Low	0x0000	false
bytes-of-machine-words-reversed-Hi	0x0000	false
media-run-from-swap	0x0000	false
network-run-from-swap	0x0000	false
general		
compiler-stamp	0x51DBA0B7	Tue Jul 09 05:33:43 2013 UTC
size-of-optional-header	0x00E0	224 bytes
signature	0x00004550	PE00
machine	0x014C	Intel-386
sections-count	0x0005	5
pointer-symbol-table	0x00000000	0x00000000
number-of-symbols	0x00000000	0x00000000

Tool Used - PEStudio

Analyzing the flagged API - there are list of APIs

API	Function	Significance
CreateProcess	Execution	Launches external programs
QueueUserWorkItem	Process Injection	Injects code via threads
MiniDumpWriteDump	Memory Dumping	Dumps process memory (e.g., for credential theft)
Sleep	Sandbox Evasion	Delays execution to evade analysis
CreateFile, ReadFile	File Operations	Creates/reads files (payload or config)
GetSystemTimeAsFileTime	Environment Check	Detects VMs or sandboxes via time checks

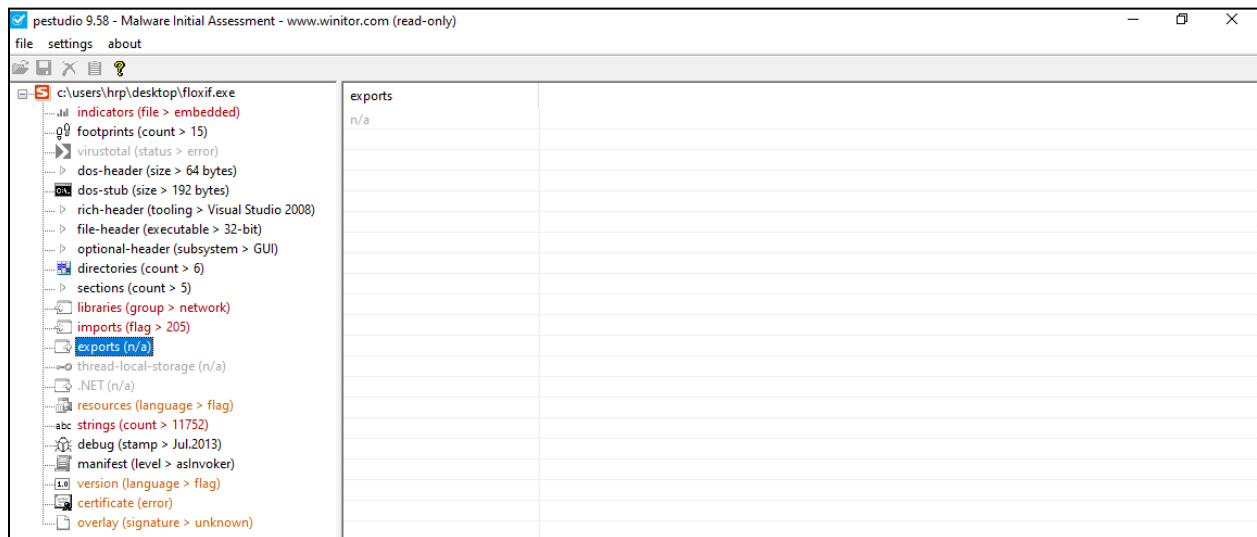
These are the screenshot of the Flagged API of the file

The screenshot shows the PEStudio 9.58 interface. The 'Imports' tab is selected, displaying a list of imported functions. The 'Imported Functions' list is expanded, showing various Windows API functions. The 'Imported Libraries' list is also visible, showing various DLLs like 'kernel32.dll', 'user32.dll', and 'GDI32.dll'. The 'Imported Functions' list is sorted by 'Technique' and 'Value'. The 'Imported Libraries' list is sorted by 'Name'.

Imported Function	Imported Library	Technique	Value
SystemTimeToFileTime	kernel32.dll	-	-
MoveFileEx	kernel32.dll	T1105 Remote File Copy	-
ReadFile	kernel32.dll	-	-
CreateDirectory	kernel32.dll	-	-
CreateFileMapping	kernel32.dll	-	-
MoveFile	kernel32.dll	T1105 Remote File Copy	-
GetFileTime	kernel32.dll	-	-
FileTimeToLocalFileTime	kernel32.dll	-	-
FileTimeToSystemTime	kernel32.dll	-	-
PathFileExists	kernel32.dll	-	-
FindFirstFile	kernel32.dll	T1083 File and Directory Discovery	-
FindClose	kernel32.dll	-	-
GetSystemTimeAsFileTime	kernel32.dll	T1124 System Time Discovery	-
GetFileType	kernel32.dll	-	-
FlushFileBuffers	kernel32.dll	-	-
CreateFile	kernel32.dll	-	-
CreateProcess	kernel32.dll	T1106 Execution through API	-
GetCurrentProcess	kernel32.dll	T1057 Process Discovery	-
GetCurrentThreadId	kernel32.dll	T1057 Process Discovery	-
ResumeThread	kernel32.dll	T1055 Process Injection	-
QueueUserWorkItem	kernel32.dll	T1055 Process Injection	-
CreateThread	kernel32.dll	-	-
SuspendThread	kernel32.dll	T1055 Process Injection	-
TlsAlloc	kernel32.dll	-	-
TlsGetValue	kernel32.dll	-	-
TlsSetValue	kernel32.dll	-	-
PostThreadMessage	kernel32.dll	-	-
ExitThread	kernel32.dll	-	-
TerminateProcess	kernel32.dll	-	-
ExitProcess	kernel32.dll	-	-
FreeEnvironmentStrings	kernel32.dll	-	-
GetEnvironmentStrings	kernel32.dll	-	-
GetCommandLine	kernel32.dll	-	-
DbgHelp.dll	kernel32.dll	-	-
MiniDumpWriteDump	kernel32.dll	-	-
Sleep	kernel32.dll	T1497 Sandbox Evasion	-

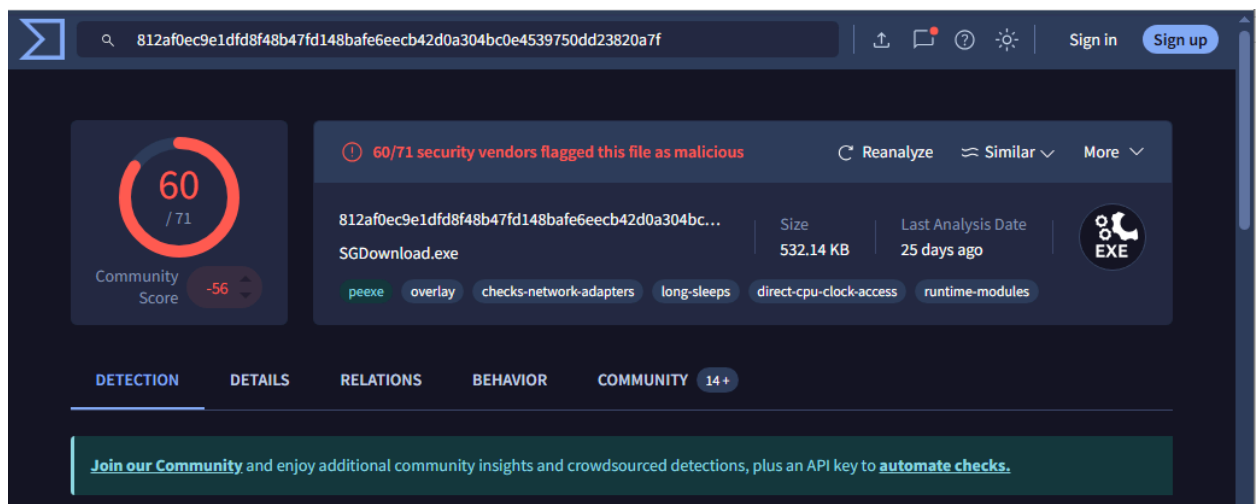
Export Table:

There is no Export data available in the file.

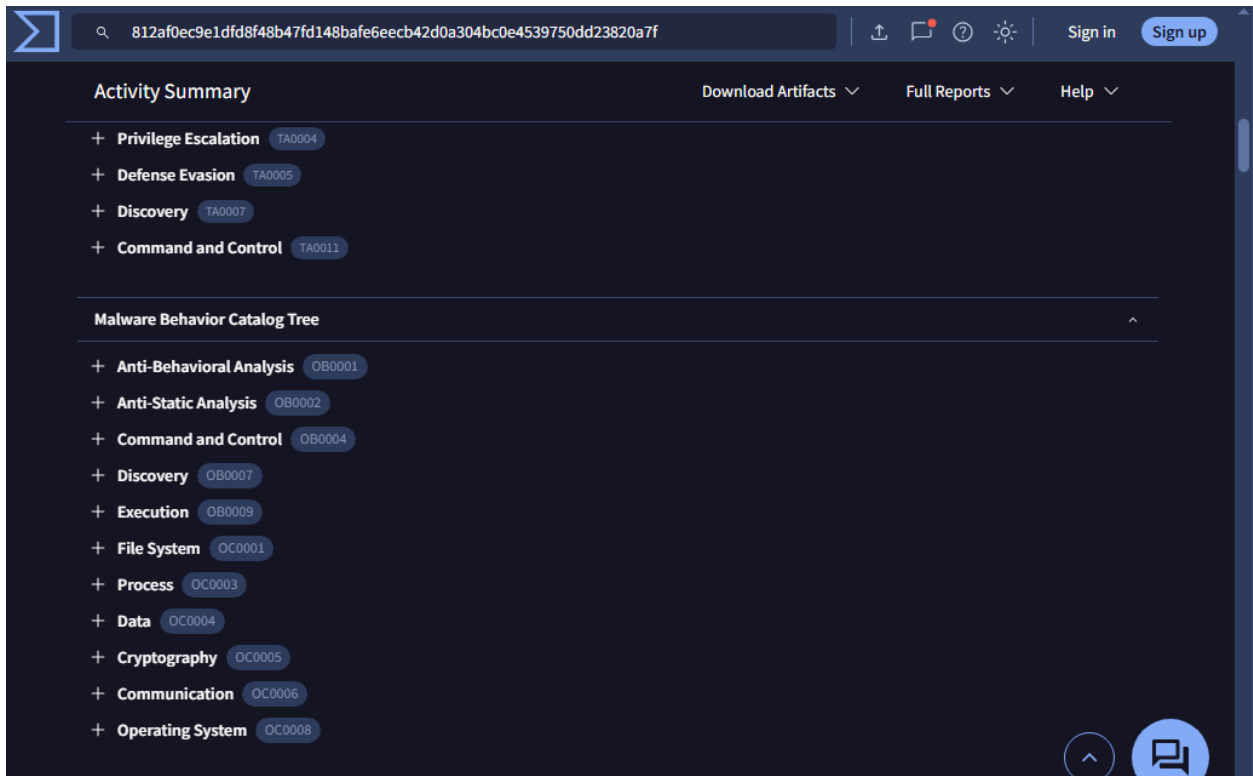


Behaviour & Analysis

Based on the [VirusTotal](#) the is a malware with some behaviours,



Behaviours:



Mapped Behaviors and API Calls

API	Function	Associated Behavior	Significance
CreateProcess	Process Execution	Execution (OB0009)	Launches external programs or malicious payloads
QueueUserWorkItem	Thread/Process Injection	Process Injection (OC0003)	Injects or schedules code execution on threads
MiniDumpWriteDump	Memory Dumping	Credential Access / Memory Dumping (OC0004)	Dumps process memory, potentially stealing credentials
Sleep	Execution Delay	Anti-Behavioral Analysis (OB0001)	Delays execution to evade sandbox or dynamic analysis

CreateFile, ReadFile	File Operations	File System Interaction (OC0001)	Creates, opens, and reads files for payload staging or configuration
GetSystemTimeAsFileTime	Environment Check	Anti-Behavioral Analysis (OB0001)	Detects sandbox or virtual environment by checking system time

Conclusion

The static analysis of the suspicious PE file using PEiD and PESTudio indicates that the binary is most likely unpacked and reveals a rich set of imports that point toward malicious capabilities. The flagged APIs highlight functions associated with execution of external processes, potential process injection, credential dumping, file system operations, and anti-analysis techniques. While no exports are present—consistent with an EXE rather than a DLL—the observed imports suggest that the sample is designed to act as a backdoor or infostealer, capable of staging additional payloads, harvesting sensitive information, and evading detection. Overall, the malware exhibits behaviors commonly associated with credential theft, persistence, and possible command-and-control communication, making it a high-risk threat.